

Determination of the Quantum Yield of the Ferrioxalate Actinometer with Electrically Calibrated Radiometers

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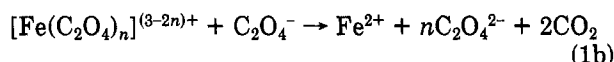
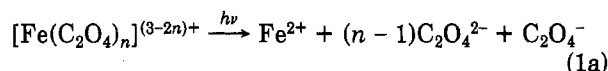
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Amplitude-stabilized laser sources and electrically calibrated radiometers were used for the absolute calibration of the quantum yield $\phi_{Fe^{2+}}$ of the ferrioxalate actinometer at three laser lines. Improved accuracy and precision were obtained compared to previously reported values. The values of $\phi_{Fe^{2+}}$ were found to be 0.845 ± 0.011 at 457.9 nm (0.15 M ferrioxalate concentration), 1.188 ± 0.012 at 406.7 nm (0.006 M) and 1.283 ± 0.023 at 363.8 nm (0.006 M). The total uncertainties are at the 99.5% confidence limits and include components for the random measurement error and the estimated bias of the measurement. The $\phi_{Fe^{2+}}$ values at 368.8 and 406.7 nm agree reasonably well with previously reported values; however, the value at 457.9 nm is $\sim 13\%$ lower than an interpolation between previously reported values. The $\phi_{Fe^{2+}}$ value at 457.9 nm was independent of the order of reagent addition, the presence of oxygen, flux density changes, and concentration of Fe^{2+} formed. Possible causes for the discrepancy at 457.9 nm are discussed. The ferrioxalate actinometer is recommended as a transfer standard for the calibration of laser power meters.

Introduction

Chemical actinometers are extensively used to measure photon fluxes. The most popular have included the ferrioxalate (200–546 nm),^{1–4} the Reineckate (350–670 nm),⁵ and, to some extent, the uranyl oxalate⁶ actinometers. More recently, a quantum flat photooxidation actinometer based on the tris(2,2'-bipyridine)ruthenium(II) cation has been proposed.⁷ The most widely used, however, is the ferrioxalate actinometer because of its sensitivity, ease of use, and relatively wide spectral range that includes the ultraviolet.

For the ferrioxalate actinometer the production of iron(II) ions proceeds by the following reactions:



Normally, the moles of Fe^{2+} formed are determined spectrophotometrically by development with 1,10-phenanthroline (phen) to form the red $[Fe(phen)_3]^{2+}$ moiety ($\lambda_{max} = 510$ nm). For wavelengths less than 436 nm, the photochemical yield $\phi_{Fe^{2+}}$ for Fe^{2+} production exceeds unity because of the secondary thermal reaction 1b.

In the calibration of a photon source, it is assumed that $\phi_{Fe^{2+}}$ is accurately known. For the ferrioxalate actinometer, the generally accepted quantum yields have been those obtained in the classic work of Hatchard and Parker.¹ These authors carried out an exhaustive study of interferences, spectral sensitivity, photon flux dependence, and the effect of ferrioxalate concentration on the quantum yield. The absolute value of their determinations of $\phi_{Fe^{2+}}$ were based on either a calibrated thermopile traceable to the National Physical Laboratory (NPL) or the quantum yields of the uranyl oxalate actinometer.⁶ They also used a thermopile as a relative standard, normalizing all of the

data to the quantum yield of the 0.006 M concentration ferrioxalate actinometer at 436 nm. Lee and Seliger² subsequently carefully reexamined the ferrioxalate actinometer at 365/6 nm using a calibrated thermopile traceable to the National Bureau of Standards (NBS) and found excellent agreement with the Hatchard and Parker NPL thermopile based values but not the Hatchard and Parker uranyl oxalate or relative thermopile based values.

This previous work was performed by using the emission lines from mercury and cadmium arc lamps as photon sources. With the wide availability and use of lasers as light sources, it has become desirable to calibrate the ferrioxalate actinometer at additional wavelengths (e.g., for the Ar^+ laser lines at 457.9, 472.7, and 514.5 nm). Preliminary attempts to calibrate the ferrioxalate actinometer using an absolute thermopile suggested that the accepted $\phi_{Fe^{2+}}$ values in the 460–510-nm region might be in error by as much as 15%.⁸

Recent developments in the technology and characterization of electrically calibrated radiometers (ECR's) at several radiometric laboratories including the NBS have led to substantial improvements in radiant power measurement accuracy.^{9–11} Radiometers of this type are absolute detectors independent of any other radiometric measurement basis, such as the blackbody (thermodynamic temperature based) absolute radiation source. Since an actinometer is a radiation detector, its calibration by

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an ECR absolute detector is a rather straightforward measurement directly traceable to electrical standards. In addition, the precision of this calibration is greatly improved by the use of amplitude-stabilized CW lasers developed by the NBS group.¹² A cooperative study was therefore initiated to reexamine the $\phi_{\text{Fe}^{2+}}$ values of the ferrioxalate actinometer using this state of the art technology.

In this paper we report the absolute quantum yields of the ferrioxalate system at the 363.8-, 406.7-, and 457.9-nm laser lines. The results show that, at the 457.9-nm line, there is a discrepancy of $\sim 13\%$ between our measurement of $\phi_{\text{Fe}^{2+}}$ and an interpolation of the previously accepted values. An inaccuracy at 457.9 nm translates directly into errors in quantum yields of other systems that were determined by using the ferrioxalate actinometer at longer wavelengths. Much smaller differences that are within the combined estimated uncertainties were obtained at the other two wavelengths. However, severe problems were noted with the reproducibility of $\phi_{\text{Fe}^{2+}}$ at 363.8 nm in concentrated ferrioxalate solutions.

Experimental Section¹³

Chemicals. Unless otherwise noted, all chemicals used were of reagent-grade quality. Potassium ferrioxalate (recrystallized 3 times from hot water)¹⁴ and the solutions of it, 1,10-phenanthroline (Eastman Lot No. B3-A, ACS grade), and buffer were freshly prepared as previously described.¹ The 1,10-phenanthroline solutions were kept in the dark and used within 3 days of preparation to avoid degradation problems.¹⁵ The specific molar absorbance ϵ at 510 nm for the $[\text{Fe}(\text{phen})_3]^{2+}$ ion was determined by the ASTM procedure E-34¹⁶ using electrolytic iron, NBS Standard Reference Material (SRM) 365.¹⁷

Spectrophotometry. Spectrophotometric determinations of the Fe^{2+} formed during the photolysis were made by using Cary 16 or 219 spectrophotometers. The wavelength scale of the instrument was calibrated by using mercury and deuterium line sources, and the absorbance scale was calibrated before and after the actinometric measurements by using NBS SRM 930d glass filters for spectrophotometry.¹⁸ These filters also were calibrated on the NBS high-accuracy spectrophotometer¹⁹ before and after the actinometric measurements. Absorbance measurements were made at 510.0 nm ($25.0 \pm 0.1^\circ\text{C}$). The specific molar absorbance measured by four determinations for the $[\text{Fe}(\text{phen})_3]^{2+}$ ion was $11110 \pm 27 \text{ L cm}^{-1} \text{ mol}^{-1}$, in good agreement with previously reported values of $\sim 11100 \text{ L cm}^{-1} \text{ mol}^{-1}$.^{3,4}

Photolysis Instrumentation. Two types of ECR's have been developed at NBS and were available for this work. One is based on a thermopile detector¹¹ and the other on a pyroelectric thermal detector.⁹ The accuracy of an ECR

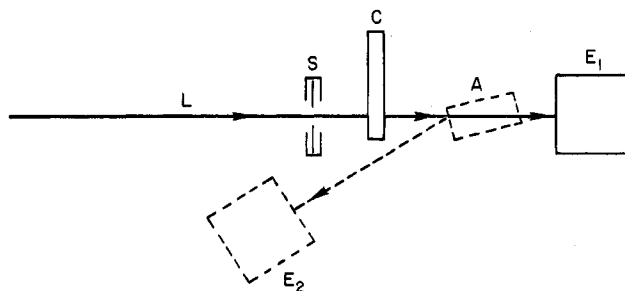


Figure 1. Schematic of actinometer calibration: (A) actinometer cell; (C) chopper; (E₁) radiometer in the photon flux measurement position; (E₂) radiometer in the reflectance measurement position; (L) amplitude-stabilized laser; (S) shutter.

is evaluated by characterizing the differences between the detector output generated in the electrical heating mode vs. that of the radiant heating mode of operation. The thrust of ECR technology has been to minimize and characterize these differences, so as to permit more accurate radiant power measurements. Because the thermopile based ECR can be more accurately characterized, it can be used to measure radiant power more accurately than the pyroelectric device; however, the pyroelectric ECR has the advantages of a faster response, greater sensitivity, and greater ease of use. Thus the pyroelectric ECR was the detector of choice for the actinometer calibration.

Two pyroelectric ECR's were used at 457.9 nm. One was completely characterized for absolute response and the other was calibrated against a thermopile ECR. No significant differences in $\phi_{\text{Fe}^{2+}}$ values were observed with either pyroelectric ECR. For the measurements at 363.8 and 406.7 nm only the latter pyroelectric ECR was used.

The laser beam amplitude stabilization system has been described elsewhere.¹² The radiant power (photon flux) was measured before and after a photolysis run and found to vary typically by less than 0.3%. These fluctuations were due to the variability of the pyroelectric ECR measurements and the residual instabilities in the laser amplitude.

A 20-W (all lines) Ar^+ laser equipped with an intracavity Littrow prism was used to produce the 457.9-nm line. For the 363.8-nm line, the intracavity prism and output mirror were replaced with UV reflecting optics. The 363.8-nm line was then isolated from the remaining UV lines by means of an external quartz prism and an optical stop. A Kr^+ laser with an intracavity Littrow prism was used to produce the 406.7-nm radiation. Plasma and thermal radiation from the laser cavity were eliminated by using optical stops and a long optical path ($\sim 3 \text{ m}$) between the laser and actinometer cell. Stray radiation was measured by detuning the laser cavity and was found to be undetectable ($<10^{-4}$ of the primary beam).

The experimental arrangement for the photolysis experiments is shown in Figure 1 and consists of a laser photon source, an electronically operated shutter, a chopper, a 5-cm long actinometer cell, and a pyroelectric ECR. West²⁰ has warned of severe errors that can arise by assuming that laser beams reflected from plane cell windows obey Fresnel's law. The errors are due to interference effects between the front and the back of the window. Although his comments were directed principally to the case in which large reflectances occur at both faces (i.e., where both the window surfaces are in contact with air or a vacuum), we have observed smaller, but significant, interference effects when the window is in contact with a

(12) J. B. Fowler, M. A. Lind, and E. F. Zalewski, *NBS Tech. Note (U.S.)*, No. 987 (1979).

(13) In order to adequately describe materials and experimental procedures, it is occasionally necessary to identify commercial products by manufacturer's name or label. In no instances does such identification imply endorsement by the National Bureau of Standards, nor does it imply that the particular product or equipment is necessarily the best available for that purpose.

(14) All operations involving potassium ferrioxalate were carried out either in the dark or with a deep red photographic "safe" light.

(15) W. D. Bowman and J. N. Demas, *J. Phys. Chem.*, **80**, 2434 (1976).

(16) *Ann. Book ASTM Stand.*, Pt. 12, 106-7 (1975).

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(19) R. Mavrodineanu, *J. Res. Natl. Bur. Stand., Sect. A*, **76**, 405 (1972).

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liquid which reduces the reflectance at one interface. For example, although a window reflectance of $\sim 3.9\%$ is predicted at 457.9 nm for a water-filled quartz cell ($n \approx 1.466$), values of 3.23–4.72% were measured from several high-quality, nonfluorescent quartz cells. Because of the added uncertainty due to interference effects, two cells, one of quartz and one of glass, were chosen for which the windows produced near Fresnelian reflectances of $\sim 4.0\%$ (water filled). The reflectance loss was measured at each wavelength as indicated in Figure 1. As a further check the windows were cut from the glass cell to measure the internal transmittance (absorbance loss) after the actinometer calibrations were completed.

The solution was mixed during photolysis by a Teflon-covered stirring bar rotated by an air-driven magnetic stirrer. Care was taken to avoid scattering the photolysis beam by the stirring bar.

Deaeration. Nitrogen bubbling was not used because it either did not provide adequate mixing or caused a variability in the quantity of ferrioxalate solution (evaporation or splash losses) as evidenced by a poor reproducibility for $\phi_{\text{Fe}^{2+}}$. However, in one set of measurements at 457.9 nm a stream of water-vapor-saturated, ultrapure N_2 was passed through the stirred ferrioxalate solution for 7 or 15 min, a thin Teflon tube being used to effect deoxygenation. After the appropriate time, the tube was withdrawn carefully from the solution and the nitrogen was continuously passed over the solution in the neck of the cell during photolysis. Photolysis was carried out in a darkened room, and the cell was further protected from stray radiation by a light-shielding enclosure.

Flux Density. For the measurements of $\phi_{\text{Fe}^{2+}}$ at 457.9 nm, the photon fluxes were changed by more than 2 orders of magnitude by varying laser power (5–20 mW) and the beam diameter (1.5–9 mm). To avoid light piping losses, the beam did not impinge on the cell wall. Transmittance of the incident laser beam by the actinometer solution was always less than 0.01% and was considered negligible.

Procedure. Photolysis runs were begun after it was ascertained that the laser amplitude and the electronics had stabilized. The measurement procedure was as follows: The laser power was measured with the ECR in position E_1 (Figure 1). The shutter was then closed, and the chopper was stopped in the open position. The photolysis cell, containing a measured volume of the ferrioxalate solution, was placed in position, and stirring was started. Photolysis was then begun by opening the shutter. Timing was performed either manually with a stopwatch or automatically with an electronic timer. Timing was accurate to ± 0.25 s and ± 0.1 s for manual and automatic control, respectively. Exposure times varied from 60 to 600 s. The laser power was immediately remeasured after the photolysis cell was removed.

The photolyzed solution was thoroughly mixed, and from two to four aliquots (3, 5, or 7 mL) of the solution were transferred by pipet to 25.0-mL volumetric flasks (the volumetric glassware was of Class A grade). Then 5.0 mL of buffer was added to the volumetric flask with mixing, and 5.0 mL of the 1,10-phenanthroline solution was added (both volumes were dispensed within 10 s by using semiautomatic, piston-driven pipettors). This order of reagent addition was as recommended in ref 15. However, the order of reagent addition was reversed during several runs with no significant differences in $\phi_{\text{Fe}^{2+}}$. The resulting solution was mixed thoroughly and diluted to the calibrated volume with distilled water. Although previous experience showed constant absorbances from <5 to >120 min of development time,¹⁵ 30-min development time was allowed

TABLE I: Summary of Quantum-Yield Values^a $\phi_{\text{Fe}^{2+}}$ and Various Measurement Parameters

λ , nm	$K_3[\text{Fe}(\text{C}_2\text{O}_4)_3]$ concn, M	$10^6(\text{radiation flux})$, einstein $\text{min}^{-1} \text{cm}^{-2}$	$\phi_{\text{Fe}^{2+}}$
363.8	0.006	1.2	1.270 ± 0.005 (4) ^b
363.8	0.006	6.0	1.294 ± 0.010 (5)
363.8	0.150	1.2	1.210 ± 0.004 (4) ^c
363.8	0.150	6.0	1.146 ± 0.043 (5) ^c
406.7	0.006	15.0	1.188 ± 0.002 (5)
457.9	0.150	1.7	0.845 ± 0.005 (2)
457.9	0.150	3.5	0.843 ± 0.001 (2)
457.9	0.150	14.5	0.846 ± 0.009 (6) ^d
457.9	0.150	23.0	0.846 ± 0.010 (2) ^e
457.9	0.150	33.0	0.847 ± 0.004 (5)
457.9	0.150	280.0	0.839 ± 0.004 (4)

^a These values are results of four separate experiments over a 4-yr period. ^b Values in parentheses are the number of independent determinations. The value for each independent determination is the average of the values obtained for a minimum of two aliquots. Indicated error is the estimated standard deviation of a single measurement. ^c We do not recommend the use of 0.150 M actinometer solution in the ultraviolet. ^d Two different, freshly prepared ferrioxalate solutions were used; the two sets of data were taken 1 yr apart. ^e Ferrioxalate solution was 14 months old.

and the developed solutions were then measured at 510.0 nm on the spectrophotometer. The absorbances of the solutions for the developed aliquots generally agreed to within 0.2%. Absorbances were kept in the range 0.3–1.1 to assure spectrophotometric accuracy (calibrated range was 0.2–1.2).

The effective reflectance of the entrance window of the cell in the photolysis experiments was measured by comparing the full beam power to the reflected beam power with the ECR. The cell was filled with ferrioxalate solution in these measurements to avoid back window reflections while maintaining the same refractive index as in the photolysis experiment. The total transmittance of the front glass cell window was measured on the Cary 16 after removal of the window from the cell. Values of the measured index of refraction were used to compute the internal transmittance of the cell window. The total forward and backward scattering from the cell window was measured and found to be negligible.

Calculations. The absolute quantum yield $\phi_{\text{Fe}^{2+}}$ was calculated by

$$\phi_{\text{Fe}^{2+}} = \frac{\eta_{\text{Fe}^{2+}}}{I_{\text{abs}}} = \frac{AV_d V_p}{(8.359 \times 10^{-9})\epsilon l V_a P \lambda t (1 - r)\tau} \quad (2)$$

where $\eta_{\text{Fe}^{2+}}$ is the moles of Fe^{2+} produced photochemically; I_{abs} is the absorbed photon dose in einsteins; V_d , V_p , and V_a are the final developed, photolyzed, and aliquot volumes, respectively (all in liters); A is the developed absorbance at 510.0 nm; ϵ is the specific molar absorbance for $[\text{Fe}(\text{phen})_3]^{2+}$ at 510.0 nm ($11110 \text{ L cm}^{-1} \text{ mol}^{-1}$); l is the spectrophotometer cell path length (1.000 cm); λ is the irradiation wavelength in nanometers; t is the irradiation time in seconds; r is the effective reflectance of the front cell window; τ is the internal transmittance of the front cell window (excluding reflection losses); and P is the average of the values of incident laser power (in watts) measured immediately before and after the photolysis.

Results and Discussion

A summary of the $\phi_{\text{Fe}^{2+}}$ values as a function of wavelength, incident photon flux, and ferrioxalate concentration is given in Table I, and a summary of the $\phi_{\text{Fe}^{2+}}$ values as a function of deoxygenation and $[\text{Fe}^{2+}]$ formed is given in

TABLE II: Summary of Quantum-Yield Values $\phi_{\text{Fe}^{2+}}$ at 457.9 nm and Various Measurement Parameters Using the 0.15 M Ferrioxalate Actinometer^a

run	photolysis time, min	soln deaeration	$10^4(\text{Fe}^{2+} \text{ produced}), \text{mol}$	$\phi_{\text{Fe}^{2+}}$
1	1	air	1.315	0.852
2	3	air	4.526	0.844
3	10	air	13.097	0.844
4	3 ^b	N ₂	4.212	0.846
5	3 ^c	N ₂	3.820	0.850

^a Three or four aliquots were taken from these runs, and the order of reagent addition was reversed for at least one aliquot from each run. The volume irradiated was 25 mL for runs 1–3, and 22 mL for runs 4 and 5. ^b N₂ bubbled for 7 min. ^c N₂ bubbled for 15 min.

TABLE III: Comparison of the $\phi_{\text{Fe}^{2+}}$ Values for the Ferrioxalate Actinometer

$K_3[\text{Fe}(\text{C}_2\text{O}_4)_3]$ concn, M	$\phi_{\text{Fe}^{2+}}$	
	this work ^a	previous work
0.006	1.283 ± 0.023 (363.8 nm)	$1.26^b, 1.26 \pm 0.03^c$ (365/366 nm)
0.006	1.188 ± 0.012 (406.7 nm)	1.16^b (405 nm)
0.15		1.01^d (436 nm)
0.15	0.845 ± 0.011 (457.9 nm)	
0.15		0.92^d (468 nm)
0.15		0.93^d (480 nm)

^a The stated uncertainties are limits of error at the 99.5% confidence level for the random measurement error and includes a component for the estimated bias of the measurement. ^b Reference 1, NPL thermopile calibration. ^c Reference 2. ^d Reference 1, thermopile used as a relative standard (relative to $\phi_{\text{Fe}^{2+}}$ of 0.006 M solution at 436 nm) and in an irradiation geometry different from that of the NPL thermopile calibrations.

Table II. The new $\phi_{\text{Fe}^{2+}}$ values recommended by this study along with relevant literature values are given in Table III. The results from our major investigation at the 457.9-nm Ar⁺ laser line show that the $\phi_{\text{Fe}^{2+}}$ values are independent of photon flux changes of more than 2 orders of magnitude (Table I) and that deoxygenation, reagent addition order, and amount of Fe²⁺ formed during photolysis also do not affect the $\phi_{\text{Fe}^{2+}}$ value (Table II). At high ferrioxalate concentration, oxidation of Fe²⁺ to Fe³⁺ does occur in weakly acidic solutions at $\sim 0.7\%$ /min, and low values of $\phi_{\text{Fe}^{2+}}$ could be obtained if excess time passes after buffer addition but before phenanthroline addition. However, the time required for the additions of both the buffer and phen solutions using the positive displacement pumps is less than 10 s, and any oxidation of the Fe²⁺ would not exceed 0.1%, a value well within our experimental error. Although most of the data were obtained by using freshly prepared 0.15 M ferrioxalate solutions (<3 weeks old), one pair of measurements made on 14-month-old 0.15 M ferrioxalate solutions gave an average $\phi_{\text{Fe}^{2+}}$ that was in excellent agreement with the $\phi_{\text{Fe}^{2+}}$ obtained by using fresh solutions.

Attempts to use a 14-month-old 0.15 M ferrioxalate solution at 363.8 nm were totally unsuccessful. Quantum yields were extremely erratic and ranged from 0.4 to 0.9—much less than the expected values of 1.20. We have even observed erratic yields when using freshly prepared 0.15 M solutions at 363.8 nm, although of a much smaller range. We also found that use of old 0.006 M ferrioxalate at 363.8 nm gave yields $\sim 3\%$ below those obtained by using fresh solutions. This irreproducibility is not unique. Severe irreproducibility was observed at 254 nm with “old”

0.006 M ferrioxalate.²¹ The source of this irreproducibility is unknown; on the basis of an observed reduced blank upon aging of the ferrioxalate solution, the opposite effect would have been expected. The fact that the problem is more severe at short wavelengths, high ferrioxalate concentrations, and high photon fluxes suggests, however, that bimolecular reactions of photochemical transients are involved.

We therefore strongly recommend that freshly prepared 0.006 M ferrioxalate solutions be used in the ultraviolet region in preference to aged 0.006 or 0.15 M solutions.

Good agreement is noted upon comparison of the 363.8- and 406.7-nm yield data with earlier literature values. In this comparison we have disregarded the $\phi_{\text{Fe}^{2+}}$ values obtained by Hatchard and Parker using the uranyl oxalate actinometer since it is a transfer measurement rather than an absolute one and the repeatability of this particular measurement is nearly 10 times worse than the one based on the NPL thermopile. Further, the results of Lee and Seliger² at 365/366 nm and our present results at 363.8 nm show the uranyl oxalate calibration to have substantial error. Our results also show the previously observed increase of $\phi_{\text{Fe}^{2+}}$ with decreasing ferrioxalate concentration at 363.8 nm.

Our recommended value of $\phi_{\text{Fe}^{2+}}$ at 457.9 nm is $\sim 13\%$ lower than the 0.97 value obtained as shown in Figure 2 by a linear (arbitrary) interpolation from the data of Hatchard and Parker. This difference is significant in the interpretation of many experiments that are based on the ferrioxalate actinometer.

It is unlikely that a decrease in the quantum yield at 457.9 nm occurs due to absorption by another species such as a dimer, since no features were observed in the absorption spectrum at this wavelength (0.006 and 0.15 M solutions;¹ and 0.25 M solution⁸). The only other explanation of the discrepancy is an error in the radiometry of the calibration. Since our $\phi_{\text{Fe}^{2+}}$ values at the three wavelengths were obtained by using exactly the same measurement technique, a systematic error occurring at 457.9 nm would also be expected at 363.8 and 406.7 nm. This was not observed.

Besides the uranyl oxalate based measurements which were discussed above, Hatchard and Parker used two other radiometric measurement procedures. These two data sets are plotted in Figure 2. At 436, 405, and 366 nm, measurements were made by using an NPL calibrated thermopile. All yield measurements were made in a uniform radiation field that overfilled the thermopile. Since the field was ascertained to be uniform, the procedure to compare photon flux density within the aperture of the thermopile ($\sim 0.4 \text{ cm}^2$) to that of the actinometer cell (0.785 cm^2) was correct. In the second case, an arc source was focused on the entrance slit of a monochromator to provide tunable, monochromatic photolysis radiation. The radiation exiting from the monochromator was then focused entirely onto the thermopile surface by means of a front surfaced mirror. In this geometry, the thermopile was calibrated at 436 nm by using the previously calibrated ferrioxalate (0.006 M) actinometer. From their data the random measurement uncertainty arising from the thermopile calibration which should be included for the subsequent measurements of $\phi_{\text{Fe}^{2+}}$ is $\sim 4.5\%$ (99.5% confidence interval). This is shown as dashed extensions to the error bars in Figure 2. This variability accounts for a portion of the discrepancy between our value and theirs.

A large error may have been introduced by their assumption that the thermopile calibration factor is valid

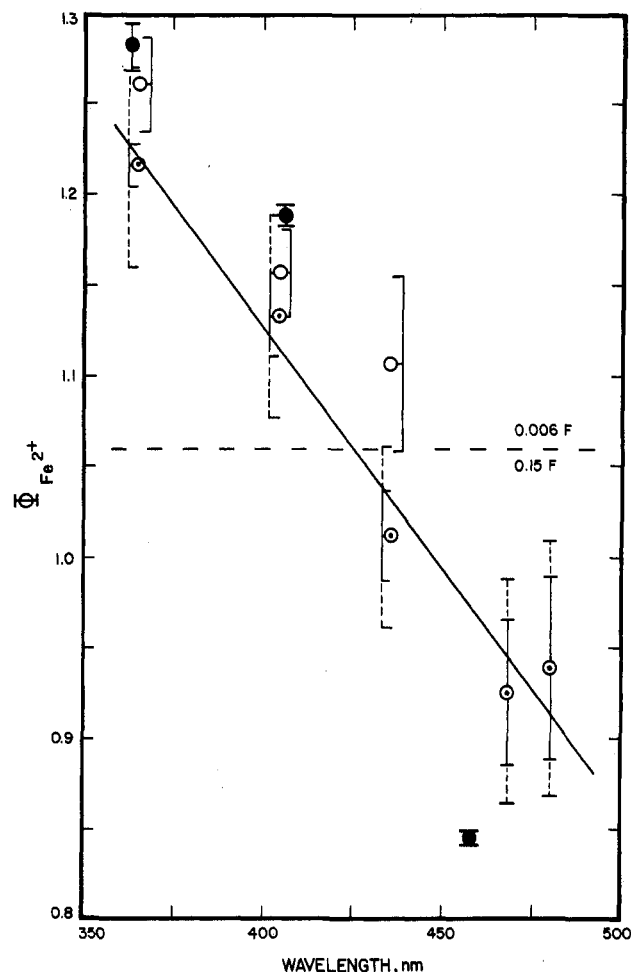


Figure 2. Quantum yield, $\Phi_{Fe^{2+}}$, as a function of wavelength: (●) this work; (○) Hatchard and Parker measurements based on an NPL calibrated thermopile; (○) Hatchard and Parker measurements based on a thermopile calibrated relative to $\Phi_{Fe^{2+}}$ at 436 nm (0.006 M). The straight-line interpolation is a fit to these data (○). The solid error bars indicate the 99.5% confidence interval, and the dashed extensions indicate the effect of the variability of the calibration based on $\Phi_{Fe^{2+}}$ (436 nm, 0.006 M). All data points above the horizontal dashed line were obtained with 0.006 M ferrioxalate solutions, while those below the line were with 0.15 M solutions.

at all other wavelengths. For this to be true, the intensity distribution across the surface of the thermopile must be wavelength independent. This assumption is not strictly valid since the spatial distribution of intensities in the arc is wavelength dependent, which, in turn, results in intensity variations across the thermopile surface as a function of wavelength.²² In addition, if a chromatic lens were used between the arc and the entrance slit of the monochromator, further intensity variations as a function of wavelength would result. Because thermopile surfaces are notoriously nonuniform in their sensitivity,^{3,4,11,21} the yield measurements made at various wavelengths using this geometry cannot be accurately related to the yield at 436 nm. We attribute the difference between our $\Phi_{Fe^{2+}}$ value at 457.9 nm and the interpolated value of Hatchard and Parker in large measure to this error source plus a contribution due to the imprecision of their thermopile calibration procedure.

An error in the Hatchard and Parker value at 457.9 nm casts doubt upon the other $\Phi_{Fe^{2+}}$ values obtained by their

TABLE IV: Estimates of Systematic Error Sources in the Determination of $\Phi_{Fe^{2+}}$ for the Ferrioxalate Actinometer

source	% uncertainty
pyroelectric ECR calibration	0.6
ECR nonuniformities	0.3
cell reflectance	0.1
cell window absorbance	0.2
ϵ_{510} for $[Fe(phen)_3]^{2+}$	0.3
Fe^{2+} oxidation	0.2
$[Fe(phen)_3]^{2+}$ absorbance	0.2
quadrature sum	0.8

use of the relative thermopile calibration procedure for both longer and shorter wavelengths.

The problem with the nonuniform irradiation geometry illustrates one of the advantages of using lasers in accurate actinometer calibrations. Because the laser beams are highly collimated, all of the radiation can be collected for a radiant power measurement rather than sampling a portion of a diverging or converging beam to measure radiant power per unit area. In the latter case one must be concerned with the accuracy of the area and the precise positioning of the apertures in the radiation field, as well as the influence of the nonuniformity of the field interacting with the nonuniformity of the detector. In the case of a radiant power measurement, only the relative nonuniformities of the detector and radiant power distribution are of prime concern.

The systematic error sources in the measurements reported here and the magnitudes of these errors are summarized in Table IV. The radiometric calibration includes the quadrature addition of 0.3% uncertainty in the NBS absolute radiant power base (thermopile ECR) with an uncertainty of 0.2% in the spectral variation of reflectivity of the pyroelectric ECR coating. To this result is arithmetically added 0.2%, which is thrice the standard deviation of the mean of the comparison of the pyroelectric ECR to the thermopile ECR. The estimated values for the other error sources of cell window reflectance and absorbance, specific molar absorbance, Fe^{2+} oxidation during color development, and $[Fe(phen)_3]^{2+}$ absorbance measurements at 510.0 nm are added to the radiometric calibration uncertainty in quadrature to give a total uncertainty in the estimated systematic error sources (bias) of 0.8%. The total uncertainty for the $\Phi_{Fe^{2+}}$ value at 457.9 nm calculated by adding the systematic error of 0.8% and random measurement uncertainty of 0.5% (99.5% confidence level for $n = 21$) is 1.3%. At 406.7 nm the random measurement uncertainty is 0.2% (99.5% confidence level, $n = 5$) for a total of 1.0%; and at 363.8 nm for the 0.006 M solution, it is 1.0% (99.5% confidence level, $n = 9$) for a total uncertainty of 1.8%. From these considerations, we estimate the overall uncertainty to be no more than 1.8% in our quantum-yield values. Indeed, we believe that the accuracy and the precision of the radiometric and spectrophotometric procedures are better than the reliability of the ferrioxalate photochemistry. Further improvement in the $\Phi_{Fe^{2+}}$ values will depend on in-depth studies to uncover and then to reduce these sources of irreproducibility.

Conclusion

We have calibrated the quantum yield of the ferrioxalate actinometer at three laser lines with improved precision and accuracy compared to previously reported values. In doing this we have demonstrated that electrical substitution radiometry represents a new and extremely accurate approach for photon flux measurements in the absolute

(22) C. A. Parker, related in a private communication that the image of the arc was defocused at the entrance slit of the monochromator which should have minimized this problem.

calibration of chemical actinometers.

In addition, the utility of chemical actinometers for measuring the intensities of high-power lasers is readily recognized and recommended. Our calibration of ferrioxalate, especially at the 457.9-nm Ar⁺ laser line, now supplies a calibration that easily provides direct laser power measurements and detector calibrations with achievable uncertainties on the order of 2%. This accuracy is difficult to surpass except with the very best silicon detector calibrations and electrical substitution radiometers. We thus recommend the use of the ferrioxalate actinometer as a transfer standard for the calibration of other types of laser radiometers.

Future work on the ferrioxalate actinometer should be

concerned with the determination of $\phi_{\text{Fe}^{2+}}$ at other useful wavelengths and establishing the reasons for the sometimes erratic behavior of the actinometer in the UV region.

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Electron Acceptor Properties of Excited *trans*-Thioindigo

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Electron transfer reactions between electron donor molecules and *trans*-thioindigo excited into its S₁ state were studied in polar and in nonpolar solvents. These investigations have shown that excited thioindigo is a strong electron acceptor so that even toluene is oxidized. In nonpolar solvents exciplexes are formed in equilibrium with the excited monomer. The investigations indicated that two emitting states of the thioindigo monomer were involved in the electron transfer process.

Introduction

In a previous paper¹ we reported on the rather unusual temperature dependence of the fluorescence quantum yield and the lifetime of *trans*-thioindigo in a nonpolar solvent consisting of a mixture of methylcyclohexane (MCH)/toluene (T) (1:1). We found a strong temperature dependence of the quantum yield, whereas the lifetime was nearly temperature independent. We interpreted the anomaly as evidence for an equilibrium between two isomeric states, the *trans* and a twisted isomer.

Further studies have shown, however, that a twisted thioindigo isomer is not formed in the first excited singlet state after excitation of the *trans* isomer. The *trans* molecule remains in its planar form, decays to its ground state, or is converted into the triplet state from which twisted configurations are formed leading finally to the formation of the *cis* isomer (*trans* → *cis* isomerization).^{2,3} Twisted configurations in the S₁ state were only found upon excitation of the *cis* isomer.⁴

According to these results our original interpretation of the fluorescence quantum yield and the lifetime cannot be quite correct. In addition, we also found that the temperature dependence of the fluorescence quantum yield varies strongly with solvent. Meanwhile, Yeh and Bard⁵ have reported that thioindigo can rather easily be electrochemically reduced. Since an excited species should be even more easily reduced, the question arises whether

toluene or similar molecules could act as electron donors. Since electron donors as well as proton donors were found to quench the *trans* → *cis* photoisomerization,^{5,6} we investigated this problem in more detail.

Experimental Section

The fluorescence intensities were measured by using a monochromator in the exciting as well as emitting light beam. The emission was detected by a GaAs photomultiplier while a photodiode was used to monitor the excitation beam. The quantum efficiencies are relative to that of rhodamine 101, the quantum yield of which is taken to be unity.⁸ The laser-flash experiments were performed by using a N₂ laser. The apparatus was described elsewhere.³ The fluorescence decay was measured by a pulsed nanosecond fluorometer, System 3000 (Photochemical Research Associates Inc (PRA)). This fluorometer, based on time-correlated single photon counting, was directly connected to a computer which was used for determining the rate constants for one- or two-exponential decay of the fluorescence intensities. This technique made it possible to determine the rate constants with excellent precision.

Most solvents were reagent grade (Fluka puriss). Acetonitrile was distilled over K₂CO₃ after refluxing over P₂O₅. Thioindigo was purified by recrystallization and vacuum sublimation. The quenchers (Fluka) were used without further purification.

Results

Polarographic measurements performed in acetonitrile yielded a half-wave reduction potential for thioindigo of

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